

SOLUTION In a vector diagram (see Figure 1.27), we again represent the course and speed to be made good by a vector $\mathbf{v}$ and the velocity of the current by $\mathbf{c}$. The correct course and speed to follow are represented by the vector $\mathbf{s}$, which is obtained by computing

$$\begin{aligned}\mathbf{s} &= \mathbf{v} - \mathbf{c} \\ &= [7 \cos 30^\circ, -7 \sin 30^\circ] - [0, -3] \\ &\approx [6.06, -3.5] - [0, -3] = [6.06, -0.5].\end{aligned}$$

Thus the captain should steer course $90^\circ - \arctan(-0.5/6.06) \approx 90^\circ + 4.7^\circ = 94.7^\circ$ and should proceed at

$$\|\mathbf{s}\| \approx \sqrt{(6.06)^2 + (-0.5)^2} \approx 6.08 \text{ knots.}$$

Proofs of the Schwarz and Triangle Inequalities

The proofs of the Schwarz and triangle inequalities illustrate the use of algebraic properties of the dot product in proving properties of the norm. Recall Eq. (11): for a vector $\mathbf{v}$ in $\mathbb{R}^n$, we have

$$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}.$$

THEOREM 1.4 Schwarz Inequality

Let $\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$. Then $|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \|\mathbf{w}\|$.

PROOF Because the norm of a vector is a real number and the square of a real number is nonnegative, for any scalars r and s we have

$$\|r\mathbf{v} + s\mathbf{w}\|^2 \geq 0. \quad (12)$$

Using relation (11), we find that

$$\begin{aligned}\|r\mathbf{v} + s\mathbf{w}\|^2 &= (r\mathbf{v} + s\mathbf{w}) \cdot (r\mathbf{v} + s\mathbf{w}) \\ &= r^2(\mathbf{v} \cdot \mathbf{v}) + 2rs(\mathbf{v} \cdot \mathbf{w}) + s^2(\mathbf{w} \cdot \mathbf{w}) \geq 0\end{aligned}$$

for all choices of scalars r and s . Setting $r = \mathbf{w} \cdot \mathbf{w}$ and $s = -(\mathbf{v} \cdot \mathbf{w})$, the preceding inequality becomes

$$\begin{aligned}(\mathbf{w} \cdot \mathbf{w})^2(\mathbf{v} \cdot \mathbf{v}) - 2(\mathbf{w} \cdot \mathbf{w})(\mathbf{v} \cdot \mathbf{w})^2 + (\mathbf{v} \cdot \mathbf{w})^2(\mathbf{w} \cdot \mathbf{w}) \\ = (\mathbf{w} \cdot \mathbf{w})^2(\mathbf{v} \cdot \mathbf{v}) - (\mathbf{w} \cdot \mathbf{w})(\mathbf{v} \cdot \mathbf{w})^2 \geq 0.\end{aligned}$$

Factoring out $(\mathbf{w} \cdot \mathbf{w})$, we see that

$$(\mathbf{w} \cdot \mathbf{w})[(\mathbf{w} \cdot \mathbf{w})(\mathbf{v} \cdot \mathbf{v}) - (\mathbf{v} \cdot \mathbf{w})^2] \geq 0. \quad (13)$$

If $\mathbf{w} \cdot \mathbf{w} = 0$, then $\mathbf{w} = \mathbf{0}$ by the positivity property in Theorem 1.3, and the Schwarz inequality is then true because it reduces to $0 \leq 0$. If $\|\mathbf{w}\|^2 = \mathbf{w} \cdot \mathbf{w} \neq 0$,

then the expression in square brackets in relation (13) must also be nonnegative—that is,

$$(\mathbf{w} \cdot \mathbf{w})(\mathbf{v} \cdot \mathbf{v}) - (\mathbf{v} \cdot \mathbf{w})^2 \geq 0,$$

and so

$$(\mathbf{v} \cdot \mathbf{w})^2 \leq (\mathbf{v} \cdot \mathbf{v})(\mathbf{w} \cdot \mathbf{w}) = \|\mathbf{v}\|^2 \|\mathbf{w}\|^2.$$

Taking square roots, we obtain the Schwarz inequality. $\blacktriangle$

The Schwarz inequality can be used to prove the triangle inequality that was illustrated in Figure 1.22.

THEOREM 1.5 The Triangle Inequality

Let $\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$. Then $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$.

PROOF Using properties of the dot product, as well as the Schwarz inequality, we have

$$\begin{aligned} \|\mathbf{v} + \mathbf{w}\|^2 &= (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} + \mathbf{w}) \\ &= (\mathbf{v} \cdot \mathbf{v}) + 2(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w}) \\ &\leq (\mathbf{v} \cdot \mathbf{v}) + 2\|\mathbf{v}\| \|\mathbf{w}\| + (\mathbf{w} \cdot \mathbf{w}) \\ &= \|\mathbf{v}\|^2 + 2\|\mathbf{v}\| \|\mathbf{w}\| + \|\mathbf{w}\|^2 \\ &= (\|\mathbf{v}\| + \|\mathbf{w}\|)^2. \end{aligned}$$

The desired relation follows at once, by taking square roots. $\blacktriangle$

SUMMARY

Let $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{w} = [w_1, w_2, \dots, w_n]$ be vectors in $\mathbb{R}^n$.

1. The *norm* or *magnitude* of $\mathbf{v}$ is $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$.
2. The norm satisfies the properties given in Theorem 1.2.
3. A *unit vector* is a vector of magnitude 1.
4. The *dot product* of $\mathbf{v}$ and $\mathbf{w}$ is $\mathbf{v} \cdot \mathbf{w} = v_1 w_1 + v_2 w_2 + \dots + v_n w_n$.
5. The dot product satisfies the properties given in Theorem 1.3.
6. Moreover, we have $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$ and $|\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\| \|\mathbf{w}\|$ (*Schwarz inequality*), and also $\|\mathbf{v} + \mathbf{w}\| \leq \|\mathbf{v}\| + \|\mathbf{w}\|$ (*triangle inequality*).
7. The *angle* θ between the vectors $\mathbf{v}$ and $\mathbf{w}$ can be found by using the relation $\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \|\mathbf{w}\| (\cos \theta)$.
8. The vectors $\mathbf{v}$ and $\mathbf{w}$ are *orthogonal* (*perpendicular*) if $\mathbf{v} \cdot \mathbf{w} = 0$.

EXERCISES

In Exercises 1–17, let $\mathbf{u} = [-1, 3, 4]$, $\mathbf{v} = [2, 1, -1]$, and $\mathbf{w} = [-2, -1, 3]$. Find the indicated quantity.

1. $\|-\mathbf{u}\|$
2. $\|\mathbf{v}\|$
3. $\|\mathbf{u} + \mathbf{v}\|$
4. $\|\mathbf{v} - 2\mathbf{u}\|$
5. $\|3\mathbf{u} - \mathbf{v} + 2\mathbf{w}\|$
6. $\left\|\frac{4}{5}\mathbf{w}\right\|$
7. The unit vector parallel to $\mathbf{u}$, having the same direction
8. The unit vector parallel to $\mathbf{w}$, having the opposite direction
9. $\mathbf{u} \cdot \mathbf{v}$
10. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w})$
11. $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w}$
12. The angle between $\mathbf{u}$ and $\mathbf{v}$
13. The angle between $\mathbf{u}$ and $\mathbf{w}$
14. The value of x such that $[x, -3, 5]$ is perpendicular to $\mathbf{u}$
15. The value of y such that $[-5, y, 10]$ is perpendicular to $\mathbf{u}$
16. A nonzero vector perpendicular to both $\mathbf{u}$ and $\mathbf{v}$
17. A nonzero vector perpendicular to both $\mathbf{u}$ and $\mathbf{w}$

In Exercises 18–21, use properties of the dot product and norm to compute the indicated quantities mentally, without pencil or paper (or calculator).

18. $\| [42, 14] \|$
19. $\| [10, 20, 25, -15] \|$
20. $[14, 21, 28] \cdot [4, 8, 20]$
21. $[12, -36, 24] \cdot [25, 30, 10]$
22. Find the angle between $[1, -1, 2, 3, 0, 4]$ and $[7, 0, 1, 3, 2, 4]$ in $\mathbb{R}^6$.
23. Prove that $(2, 0, 4)$, $(4, 1, -1)$, and $(6, 7, 7)$ are vertices of a right triangle in $\mathbb{R}^3$.

24. Prove that the angle between two unit vectors $\mathbf{u}_1$ and $\mathbf{u}_2$ in $\mathbb{R}^n$ is $\arccos(\mathbf{u}_1 \cdot \mathbf{u}_2)$.

In Exercises 25–30, classify the vectors as parallel, perpendicular, or neither. If they are parallel, state whether they have the same direction or opposite directions.

25. $[-1, 4]$ and $[8, 2]$
26. $[-2, -1]$ and $[5, 2]$
27. $[3, 2, 1]$ and $[-9, -6, -3]$
28. $[2, 1, 4, -1]$ and $[0, 1, 2, 4]$
29. $[10, 4, -1, 8]$ and $[-5, -2, 3, -4]$
30. $[4, 1, 2, 1, 6]$ and $[8, 2, 4, 2, 3]$
31. The distance between points $(v_1, v_2, \dots, v_n)$ and $(w_1, w_2, \dots, w_n)$ in $\mathbb{R}^n$ is the norm $\|\mathbf{v} - \mathbf{w}\|$, where $\mathbf{v} = [v_1, v_2, \dots, v_n]$ and $\mathbf{w} = [w_1, w_2, \dots, w_n]$. Why is this a reasonable definition of distance?

In Exercises 32–35, use the definition given in Exercise 31 to find the indicated distance.

32. The distance from $(-1, 4, 2)$ to $(0, 8, 1)$ in $\mathbb{R}^3$
33. The distance from $(2, -1, 3)$ to $(4, 1, -2)$ in $\mathbb{R}^3$
34. The distance from $(3, 1, 2, 4)$ to $(-1, 2, 1, 2)$ in $\mathbb{R}^4$
35. The distance from $(-1, 2, 1, 4, 7, -3)$ to $(2, 1, -3, 5, 4, 5)$ in $\mathbb{R}^6$
36. The captain of a barge wishes to get to a point directly across a straight river that runs from north to south. If the current flows directly downstream at 5 knots and the barge steams at 13 knots, in what direction should the captain steer the barge?
37. A 100-lb weight is suspended by a rope passed through an eyelet on top of the weight and making angles of 30° with the vertical, as shown in Figure 1.28. Find the tension (magnitude of the force vector) along the rope. [HINT: The sum of the force vectors

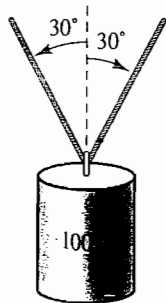


FIGURE 1.28
Both halves of the rope make an angle of 30° with the vertical.

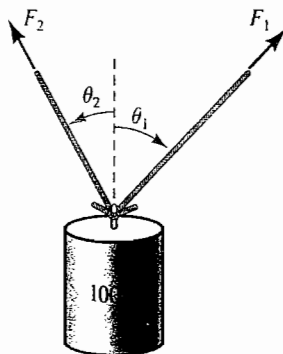


FIGURE 1.29
Two ropes tied at the eyelet and making angles θ_1 and θ_2 with the vertical.

along the two halves of the rope at the eyelet must be an upward vertical vector of magnitude 100.]

38. a. Answer Exercise 37 if each half of the rope makes an angle of θ with the vertical at the eyelet.
b. Find the tension in the rope if both halves are vertical ($\theta = 0$).
c. What happens if an attempt is made to stretch the rope out straight (horizontal) while the 100-lb weight hangs on it?
39. Suppose that a weight of 100 lb is suspended by two different ropes tied at an eyelet on top of the weight, as shown in Figure 1.29. Let the angles the ropes make with the vertical be θ_1 and θ_2 , as shown in the figure. Let the tensions in the ropes be T_1 for the right-hand rope and T_2 for the left-hand rope.
- Show that the force vector $\mathbf{F}_1$ shown in Figure 1.29 is $T_1(\sin \theta_1)\mathbf{i} + T_1(\cos \theta_1)\mathbf{j}$.
 - Find the corresponding expression for $\mathbf{F}_2$ in terms of T_2 and θ_2 .
 - If the system is in equilibrium, $\mathbf{F}_1 + \mathbf{F}_2 = 100\mathbf{j}$, so $\mathbf{F}_1 + \mathbf{F}_2$ must have $\mathbf{i}$ -component 0 and $\mathbf{j}$ -component 100. Write two equations reflecting this fact, using the answers to parts (a) and (b).
 - Find T_1 and T_2 if $\theta_1 = 45^\circ$ and $\theta_2 = 30^\circ$.

40. Mark each of the following True or False.
- Every nonzero vector in $\mathbb{R}^n$ has nonzero magnitude.
 - Every vector of nonzero magnitude in $\mathbb{R}^n$ is nonzero.
 - The magnitude of $\mathbf{v} + \mathbf{w}$ must be at least as large as the magnitude of either $\mathbf{v}$ or $\mathbf{w}$ in $\mathbb{R}^n$.
 - Every nonzero vector $\mathbf{v}$ in $\mathbb{R}^n$ has exactly one unit vector parallel to it.
 - There are exactly two unit vectors parallel to any given nonzero vector in $\mathbb{R}^n$.
 - There are exactly two unit vectors perpendicular to any given nonzero vector in $\mathbb{R}^n$.
 - The angle between two nonzero vectors in $\mathbb{R}^n$ is less than 90° if and only if the dot product of the vectors is positive.
 - The dot product of a vector with itself yields the magnitude of the vector.
 - For a vector $\mathbf{v}$ in $\mathbb{R}^n$, the magnitude of r times $\mathbf{v}$ is r times the magnitude of $\mathbf{v}$.
 - If $\mathbf{v}$ and $\mathbf{w}$ are vectors in $\mathbb{R}^n$ of the same magnitude, then the magnitude of $\mathbf{v} - \mathbf{w}$ is 0.
41. Prove the indicated property of the norm stated in Theorem 1.2.
- The positivity property
 - The homogeneity property

42. Prove the indicated property of the dot product stated in Theorem 1.3.
 - a. The commutative law
 - b. The distributive law
 - c. The homogeneity property
43. For vectors $\mathbf{v}$ and $\mathbf{w}$ in $\mathbb{R}^n$, prove that $\mathbf{v} - \mathbf{w}$ and $\mathbf{v} + \mathbf{w}$ are perpendicular if and only if $\|\mathbf{v}\| = \|\mathbf{w}\|$.
44. For vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in $\mathbb{R}^n$ and for scalars r and s , prove that, if $\mathbf{w}$ is perpendicular to both $\mathbf{u}$ and $\mathbf{v}$, then $\mathbf{w}$ is perpendicular to $r\mathbf{u} + s\mathbf{v}$.
45. Use vector methods to prove that the diagonals of a rhombus (parallelogram with equal sides) are perpendicular. [HINT: Use a figure similar to Figure 1.25 and one of the preceding exercises.]
46. Use vector methods to prove that the midpoint of the hypotenuse of a right triangle is equidistant from the three vertices. [HINT: See Figure 1.30. Show that

$$\left\| \frac{1}{2}(\mathbf{v} + \mathbf{w}) \right\| = \left\| \frac{1}{2}(\mathbf{v} - \mathbf{w}) \right\|$$

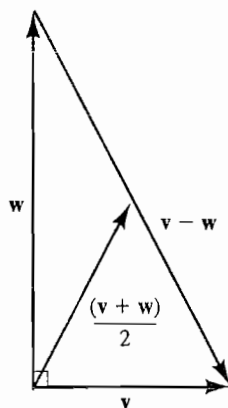


FIGURE 1.30

The vector $\frac{1}{2}(\mathbf{v} + \mathbf{w})$ to the midpoint of the hypotenuse.

MATLAB

MATLAB has a built-in function `norm(x)` for computing the norm of a vector $\mathbf{x}$. It has no built-in command for finding a dot product or the angle between two vectors. Because one purpose of these exercises is to give practice at working with MATLAB, we will show how the norm of a vector can be computed without using the built-in function, as well as how to compute dot products and angles between vectors.

It is important to know how to enter data into MATLAB. In Section 1.1, we showed how to enter a vector. We have created M-files on the LINTEK disk that can be used to enter data automatically for our exercises, once practice in manual data entry has been provided. If these M-files have been copied into your MATLAB, you can simply enter `fbc1s2` to create the data vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ for the exercises below. The name of the file containing them is `FBC1S2.M`, where the `FBC1S2` stands for "Fraleigh/Beauregard Chapter 1 Section 2." To view this data file so that you can create data files of your own, if you wish, simply enter `type fbc1s2` when in MATLAB. In addition to saving time, the data files help prevent wrong answers resulting from typos in data entry.

Access MATLAB, and either enter `fbc1s2` or manually enter the following vectors.

$$\begin{aligned} \mathbf{a} &= [-2 \ 1 \ 3 \ 5 \ 1] & \mathbf{u} &= [2/3 \ -4/7 \ 8/5] \\ \mathbf{b} &= [4 \ -1 \ 2 \ 3 \ 5] & \mathbf{v} &= [-1/2 \ 13/3 \ 17/11] \\ \mathbf{c} &= [-1 \ 0 \ 3 \ 0 \ 4] & \mathbf{w} &= [22/7 \ 15/2 \ -8/3] \end{aligned}$$

If you enter these vectors manually, be sure to proofread your data entry for accuracy. Enter `help .` to see what can be done with the period. Enter `a.*b` and compare the resulting vector with the vectors `a` and `b`. (Be sure to get in the habit of putting a space before the period so that MATLAB will never interpret it as a decimal point in this context.) Now enter `c.^3` and compare with the vector `c`. The symbol `^` is used to denote exponentiation. Then enter `sum(a)` and note how this number was obtained from the vector `a`. Using the `.` notation, the sum function `sum(x)`, the square root function `sqrt(x)`, and the arccosine function `acos(x)`, we can easily write formulas for computing norms of vectors, dot products of vectors, and the angle between two vectors.

- M1. Enter `x = a` and then enter `normx = sqrt(sum(x.*x))` to find $\|a\|$. Compare your answer with the result obtained by entering `norm(a)`.
- M2. Find $\|b\|$ by entering `x = b` and then pressing the upward arrow until equation `normx = sqrt(sum(x.*x))` is at the cursor, and then pressing the Enter key.
- M3. Using the technique outlined in Exercise M2, find $\|u\|$.
- M4. Using the appropriate MATLAB commands, compute the dot product $v \cdot w$ in (a) short format and (b) rational format.
- M5. Repeat Exercise M4 for $(2u - 3v) \cdot (4u - 7v)$.

NOTE: If you are working with your own personal MATLAB, you can add a function `angl(x, y)` for finding the angle between vectors `x` and `y` having the same number of components to MATLAB's supply of available functions. First, enter `help angl` to be sure that MATLAB does not already have a command with the name you will use; otherwise you might delete an existing MATLAB function. Then, assuming that MATLAB has created a subdirectory `C:\MATLAB\MATLAB` on your disk, get out of MATLAB and either use a word processor that will create ASCII files or skip to the next paragraph. Using a word processor, create an ASCII file designated as `C:\MATLAB\MATLAB\ANGL.M` by entering each of the following lines.

```
function z = angl(x, y)
% ANGL  ANGL (x, y) is the radian angle between vectors x and y.
z = acos(sum(x.*y)/(norm(x)*norm(y)))
```

Then save the file. This creates a function `angl(x, y)` of two variables `x` and `y` in place of the name `anglxy` we use in M6. You will now be able to compute the angle between vectors `x` and `y` in MATLAB simply by entering `angl(x, y)`. Note that the file name `ANGL.M` concludes with the `.M` suffix. MATLAB comes with a number of these M-files, which are probably in the subdirectory `MATLAB\MATLAB` of your disk. Remember to enter `help angl` from MATLAB first, to be sure there is not already a file with the name `ANGL.M`.

If you do not have a word processor that writes ASCII files, you can still create the file from DOS if your hard disk is the default c-drive. First enter `CD C:\MATLAB\MATLAB`. Then enter `COPY CON ANGL.M` and proceed to enter the three lines displayed above. When you have pressed Enter after the final line, press the F6 key and then press Enter again.

After creating the file, access MATLAB and test your function `angl(x, y)` by finding the angle between the vectors `[1, 0]` and `[-1, 0]`. The angle should be $\pi \approx 3.1416$. Then enter `help angl` and you should see the explanatory note on the line of the file that starts with `%` displayed on the screen. Using these directions as a model, you can easily create functions of your own to add to MATLAB.

M6. Enter $x = a$ and enter $y = b$. Then enter

$$\text{anglxy} = \text{acos}(\text{sum}(x .* y) / (\text{norm}(x) * \text{norm}(y)))$$

to compute the angle (in radians) between a and b . You should study this formula until you understand why it provides the angle between a and b .

M7. Compute the angle between b and c using the technique suggested in Exercise M2. Namely, enter $x = b$, enter $y = c$, and then use the upward arrow until the cursor is at the formula for anglxy and press Enter.

M8. Move the cursor to the formula for anglxy and edit the formula so that the angle will be given in degrees rather than in radians. Recall that we multiply by $180/\pi$ to convert radians to degrees. The number π is available as pi in MATLAB. Check your editing by computing the angle between the vectors $[1, 0]$ and $[0, 1]$. Then find the angle between u and w in degrees.

M9. Find the angle between $3u - 2w$ and $4v + 2w$ in degrees.

1.3

MATRICES AND THEIR ALGEBRA

The Notation $Ax = b$

We saw in Section 1.1 that we can write a linear system such as

$$\begin{aligned} x_1 - 2x_2 &= -1 \\ 3x_1 + 5x_2 &= 19 \end{aligned} \quad (1)$$

in the unknowns x_1 and x_2 as a single column vector equation—namely,

$$x_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 5 \end{bmatrix} = \begin{bmatrix} -1 \\ 19 \end{bmatrix}. \quad (2)$$

Another useful way to abbreviate this linear system is

$$\begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 19 \end{bmatrix}. \quad (3)$$

$A \qquad \qquad x \qquad \qquad b$

Let us denote by A the bracketed array on the left containing the coefficients of the linear system. This array A is followed by the column vector x of unknowns, and let the column vector of constants after the equal sign be denoted by b . We can then symbolize the linear system as

$$Ax = b. \quad (4)$$

There are several reasons why notation (4) is a convenient way to write a linear system. It is much easier to denote a general linear system by $Ax = b$ than to write out several linear equations with unknowns $x_1, x_2, \dots, x_n$, subscripted letters for the coefficients of the unknowns, and constants $b_1, b_2, \dots, b_m$ to the

right of the equal signs. [Just look ahead at Eq. (1) on page 54.] Also, a single linear equation in just one unknown can be written in the form $ax = b$ ($2x = 6$, for example), and the notation $A\mathbf{x} = \mathbf{b}$ is suggestively similar. Furthermore, we will see in Section 2.3 that we can regard such an array A as defining a *function* whose value at $\mathbf{x}$ we will write as $A\mathbf{x}$, much as we write $\sin x$. Solving a linear system $A\mathbf{x} = \mathbf{b}$ can thus be regarded as finding the vector $\mathbf{x}$ such that this function applied to $\mathbf{x}$ yields the vector $\mathbf{b}$. For all of these reasons, the notation $A\mathbf{x} = \mathbf{b}$ for a linear system is one of the most useful notations in mathematics.

It is very important to remember that

$A\mathbf{x}$ is equal to a linear combination of the *column vectors* of A ,

as illustrated by Eqs. (2) and (3)—namely,

$$\begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 5 \end{bmatrix}. \quad (5)$$

The Notion of a Matrix

We now introduce the usual terminology and notation for an array of numbers such as the coefficient array A in Eq. (3).

A *matrix* is an ordered rectangular array of numbers, usually enclosed in parentheses or square brackets. For example,

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 0 & 3 \\ 2 & 1 & 4 \\ 4 & 5 & -6 \\ -3 & -1 & -1 \end{bmatrix}$$

are matrices. We will generally use upper-case letters to denote matrices.

The size of a matrix is specified by the number of (horizontal) rows and the number of (vertical) columns that it contains. The matrix A above contains two rows and two columns and is called a 2×2 (read “2 by 2”) matrix. Similarly, B is a 4×3 matrix. In writing the notation $m \times n$ to describe the shape of a matrix, we always write the number of rows first. An $n \times n$ matrix has the same number of rows as columns and is said to be a *square matrix*. We recognize that a $1 \times n$ matrix is a row vector with n components, and an $m \times 1$ matrix is a column vector with m components. The rows of a matrix are its *row vectors* and the columns are its *column vectors*.

Double subscripts are commonly used to indicate the location of an entry in a matrix that is not a row or column vector. The first subscript gives the number of the row in which the entry appears (counting from the top), and the

second subscript gives the number of the column (counting from the left). Thus an $m \times n$ matrix A may be written as

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

If we want to express the matrix B on page 36 as $[b_{ij}]$, we would have $b_{11} = -1$, $b_{21} = 2$, $b_{32} = 5$, and so on.

Matrix Multiplication

We are going to consider the expression Ax shown in Eq. (3) to be the *product* of the matrix A and the column vector x . Looking back at Eq. (5), we see that such a product of a matrix A with a column vector x should be the linear combination of the column vectors of A having as coefficients the components in the vector x . Here is a *nonsquare* example in which we replace the vector x of unknowns by a specific vector of numbers.

EXAMPLE 1 Write as a linear combination and then compute the product

$$\begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & -7 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \\ 8 \end{bmatrix}.$$

HISTORICAL NOTE THE TERM *MATRIX* is first mentioned in mathematical literature in an 1850 paper of James Joseph Sylvester (1814–1897). The standard nontechnical meaning of this term is “a place in which something is bred, produced, or developed.” For Sylvester, then, a matrix, which was an “oblong arrangement of terms,” was an entity out of which one could form various square pieces to produce determinants. These latter quantities, formed from square matrices, were quite well known by this time.

James Sylvester (his original name was James Joseph) was born into a Jewish family in London, and was to become one of the supreme algebraists of the nineteenth century. Despite having studied for several years at Cambridge University, he was not permitted to take his degree there because he “professed the faith in which the founder of Christianity was educated.” Therefore, he received his degrees from Trinity College, Dublin. In 1841 he accepted a professorship at the University of Virginia; he remained there only a short time, however, his horror of slavery preventing him from fitting into the academic community. In 1871 he returned to the United States to accept the chair of mathematics at the newly opened Johns Hopkins University. In between these sojourns, he spent about 10 years as an attorney, during which time he met Arthur Cayley (see the note on p. 3), and 15 years as Professor of Mathematics at the Royal Military Academy, Woolwich. Sylvester was an avid poet, prefacing many of his mathematical papers with examples of his work. His most renowned example was the “Rosalind” poem, a 400-line epic, each line of which rhymed with “Rosalind.”

SOLUTION Using Eq. (5) as a guide, we find that

$$\begin{bmatrix} 2 & -3 & 5 \\ -1 & 4 & -7 \end{bmatrix} \begin{bmatrix} -2 \\ 5 \\ 8 \end{bmatrix} = -2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 5 \begin{bmatrix} -3 \\ 4 \end{bmatrix} + 8 \begin{bmatrix} 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 21 \\ -34 \end{bmatrix}.$$

Note that in Example 1, the first entry 21 of the final column vector is computed as $(-2)(2) + (5)(-3) + (8)(5)$, which is precisely the dot product of the first row vector $[2 \ -3 \ 5]$ of the matrix with the column vector $\begin{bmatrix} -2 \\ 5 \\ 8 \end{bmatrix}$.

Similarly, the second component -34 of our answer is the dot product of the second row vector $[-1 \ 4 \ -7]$ with this column vector.

In a similar fashion, we see that the i th component of a column vector $A\mathbf{b}$ will be equal to the dot product of the i th row of A with the column vector $\mathbf{b}$. We should also note from Example 1 that the number of components in a row of A will have to be equal to the number of components in the column vector $\mathbf{b}$ if we are to compute the product $A\mathbf{b}$.

We have illustrated how to compute a product $A\mathbf{b}$ of an $m \times n$ matrix with an $n \times 1$ column vector. We can extend this notion to a product AB of an $m \times n$ matrix A with an $n \times s$ matrix B .

The product AB is the matrix whose j th column is the product of A with the j th column vector of B .

Letting $\mathbf{b}_j$ be the j th column vector of B , we write $AB = C$ symbolically as

$$A \begin{bmatrix} | & | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_s \\ | & | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} | & | & | & \cdots & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_s \\ | & | & | & \cdots & | \end{bmatrix}.$$

$B \qquad C$

Because B has s columns, C has s columns. The comments after Example 1 indicate that the i th entry in the j th column of AB is the dot product of the i th row of A with the j th column of B . We give a formal definition.

DEFINITION 1.8 Matrix Multiplication

Let $A = [a_{ik}]$ be an $m \times n$ matrix, and let $B = [b_{kj}]$ be an $n \times s$ matrix. The **matrix product** AB is the $m \times s$ matrix $C = [c_{ij}]$, where c_{ij} is the dot product of the i th row vector of A and the j th column vector of B .

We illustrate the choice of row i from A and column j from B to find the element c_{ij} in AB , according to Definition 1.8, by the equation

$$AB = [c_{ij}] = \begin{bmatrix} a_{i1} & \cdots & a_{in} \\ \vdots & & \vdots \\ a_{i1} & \cdots & a_{in} \\ \vdots & & \vdots \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1s} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{ns} \end{bmatrix},$$

where

$$c_{ij} = (\text{ith row vector of } A) \cdot (\text{jth column vector of } B).$$

In summation notation, we have

$$\begin{aligned} c_{ij} &= a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \\ &= \sum_{k=1}^n a_{ik}b_{kj}. \end{aligned} \quad (6)$$

Notice again that AB is defined only when the second size-number (the number of columns) of A is the same as the first size-number (the number of rows) of B . The product matrix has the shape

$$(\text{First size-number of } A) \times (\text{Second size-number of } B).$$

EXAMPLE 2 Let A be a 2×3 matrix, and let B be a 3×5 matrix. Find the sizes of AB and BA , if they are defined.

SOLUTION Because the second size-number, 3, of A equals the first size-number, 3, of B , we see that AB is defined; it is a 2×5 matrix. However, BA is not defined, because the second size-number, 5, of B is not the same as the first size-number, 2, of A . ■

EXAMPLE 3 Compute the product

$$\begin{bmatrix} -2 & 3 & 2 \\ 4 & 6 & -2 \end{bmatrix} \begin{bmatrix} 4 & -1 & 2 & 5 \\ 3 & 0 & 1 & 1 \\ -2 & 3 & 5 & -3 \end{bmatrix}.$$

SOLUTION The product is defined, because the left-hand matrix is 2×3 and the right-hand matrix is 3×4 ; the product will have size 2×4 . The entry in the first row and first column position of the product is obtained by taking the dot product of the first row vector of the left-hand matrix and the first column vector of the right-hand matrix, as follows:

$$(-2)(4) + (3)(3) + (2)(-2) = -8 + 9 - 4 = -3.$$

The entry in the second row and third column of the product is the dot product of the second row vector of the left-hand matrix and the third column vector of the right-hand one:

$$(4)(2) + (6)(1) + (-2)(5) = 8 + 6 - 10 = 4,$$

and so on, through the remaining row and column positions of the product. Eight such computations show that

$$\begin{bmatrix} -2 & 3 & 2 \\ 4 & 6 & -2 \end{bmatrix} \begin{bmatrix} 4 & -1 & 2 & 5 \\ 3 & 0 & 1 & 1 \\ -2 & 3 & 5 & -3 \end{bmatrix} = \begin{bmatrix} -3 & 8 & 9 & -13 \\ 38 & -10 & 4 & 32 \end{bmatrix}.$$

Examples 2 and 3 show that sometimes AB is defined when BA is not. Even if both AB and BA are defined, however, it need not be true that $AB = BA$:

Matrix multiplication is not commutative.

EXAMPLE 4 Let

$$A = \begin{bmatrix} 0 & 2 \\ 3 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ 2 & 5 \end{bmatrix}.$$

Compute AB and BA .

HISTORICAL NOTE MATRIX MULTIPLICATION originated in the composition of linear substitutions, fully explored by Carl Friedrich Gauss (1777–1855) in his *Disquisitiones Arithmeticae* of 1801 in connection with his study of quadratic forms. Namely, if $F = Ax^2 + 2Bxy + Cy^2$ is such a form, then the linear substitution

$$x = ax' + by' \quad y = cx' + dy' \tag{i}$$

transforms F into a new form F' in the variables x' and y' . If a second substitution

$$x' = ex'' + fy'' \quad y' = gx'' + hy'' \tag{ii}$$

transforms F' into a form F'' in x'' , y'' , then the composition of the substitutions, found by replacing x' , y' in (i) by their values in (ii), gives a substitution transforming F into F'' :

$$x = (ae + bg)x'' + (af + bh)y'' \quad y = (ce + dg)x'' + (cf + dh)y''. \tag{iii}$$

The coefficient matrix of substitution (iii) is the product of the coefficient matrices of substitutions (i) and (ii). Gauss performed an analogous computation in his study of substitutions in forms in three variables, which produced the rule for multiplication of 3×3 matrices.

Gauss, however, did not explicitly refer to this idea of composition as a “multiplication.” That was done by his student Ferdinand Gotthold Eisenstein (1823–1852), who introduced the notation $S \times T$ to denote the substitution composed of S and T . About this notation Eisenstein wrote, “An algorithm for calculation can be based on this; it consists of applying the usual rules for the operations of multiplication, division, and exponentiation to symbolical equations between linear systems; correct symbolical equations are always obtained, the sole consideration being that the order of the factors may not be altered.”

SOLUTION We compute that

$$AB = \begin{bmatrix} 4 & 10 \\ 10 & 28 \end{bmatrix}, \text{ and } BA = \begin{bmatrix} 3 & 5 \\ 15 & 29 \end{bmatrix}.$$

Of course, for a square matrix A , we denote AA by A^2 , AAA by A^3 , and so on. It can be shown that matrix multiplication is associative; that is,

$$A(BC) = (AB)C$$

whenever the product is defined. This is not difficult to prove from the definition, although keeping track of subscripts can be a bit challenging. We leave the proof as Exercise 33, whose solution is given in the back of this text.

The $n \times n$ Identity Matrix

Let I be the $n \times n$ matrix $[a_{ij}]$ such that $a_{ii} = 1$ for $i = 1, \dots, n$ and $a_{ij} = 0$ for $i \neq j$. That is,

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} = \begin{bmatrix} 1 & & & & \\ & 1 & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{bmatrix},$$

where the large zeros above and below the diagonal in the second matrix indicate that each entry of the matrix in those positions is 0. If A is any $m \times n$ matrix and B is any $n \times s$ matrix, we can show that

$$AI = A \quad \text{and} \quad IB = B.$$

We can understand why this is so if we think about why it is that

$$\begin{bmatrix} 2 & 3 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 7 \end{bmatrix}.$$

Because of the relations $AI = A$ and $IB = B$, the matrix I is called the $n \times n$ **identity matrix**. It behaves for multiplication of $n \times n$ matrices exactly as the scalar 1 behaves for multiplication of scalars. We have one such square identity matrix for each integer $1, 2, 3, \dots$. To keep notation simple, we denote them all by I , rather than by $I_1, I_2, I_3, \dots$. The size of I will be clear from the context.

The identity matrix is an example of a **diagonal matrix**—namely, a square matrix with zero entries except possibly on the **main diagonal**, which extends from the upper left corner to lower right corner.

Other Matrix Operations

Although multiplication is a very important matrix operation for our work, we will have occasion to add and subtract matrices, and to multiply a matrix

by a scalar, in later chapters. Matrix addition, subtraction, and scalar multiplication are natural extensions of these same operations for vectors as defined in Section 1.1; they are again performed on entries in corresponding positions.

DEFINITION 1.9 Matrix Addition

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices of the same size $m \times n$. The **sum** $A + B$ of these two matrices is the $m \times n$ matrix $C = [c_{ij}]$, where

$$c_{ij} = a_{ij} + b_{ij}.$$

That is, the sum of two matrices of the same size is the matrix of that size obtained by adding corresponding entries.

EXAMPLE 5 Find

$$\begin{bmatrix} 1 & 2 & -4 \\ 0 & 3 & -1 \end{bmatrix} + \begin{bmatrix} -1 & 0 & 2 \\ 1 & -5 & 3 \end{bmatrix}.$$

SOLUTION The sum is the matrix

$$\begin{bmatrix} 0 & 2 & -2 \\ 1 & -2 & 2 \end{bmatrix}.$$

EXAMPLE 6 Find

$$\begin{bmatrix} 1 & -3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} -5 & 4 & 6 \\ 3 & 7 & -1 \end{bmatrix}.$$

SOLUTION The sum is undefined, because the matrices are not the same size. ■

Let A be an $m \times n$ matrix, and let O be the $m \times n$ matrix all of whose entries are zero. Then,

$$A + O = O + A = A.$$

The matrix O is called the $m \times n$ **zero matrix**; the size of such a zero matrix is made clear by the context.

DEFINITION 1.10 Scalar Multiplication

Let $A = [a_{ij}]$, and let r be a scalar. The **product** rA of the scalar r and the matrix A is the matrix $B = [b_{ij}]$ having the same size as A , where

$$b_{ij} = ra_{ij}.$$

EXAMPLE 7 Find

$$2 \begin{bmatrix} -2 & 1 \\ 3 & -5 \end{bmatrix}.$$

SOLUTION Multiplying each entry of the matrix by 2, we obtain the matrix

$$\begin{bmatrix} -4 & 2 \\ 6 & -10 \end{bmatrix}. \quad \blacksquare$$

For matrices A and B of the same size, we define the **difference** $A - B$ to be

$$A - B = A + (-1)B.$$

The entries in $A - B$ are obtained by subtracting the entries of B from entries in the corresponding positions in A .

EXAMPLE 8 If

$$A = \begin{bmatrix} 3 & -1 & 4 \\ 0 & 2 & -5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 0 & 5 \\ 4 & -2 & 1 \end{bmatrix},$$

find $2A - 3B$.

SOLUTION We find that

$$2A - 3B = \begin{bmatrix} 9 & -2 & -7 \\ -12 & 10 & -13 \end{bmatrix}. \quad \blacksquare$$

We introduced the *transpose operation* to change a row vector to a column vector, or vice versa, in Section 1.1. We generalize this operation for application to matrices, changing all the row vectors to column vectors, which results in all the column vectors becoming row vectors.

DEFINITION 1.11 Transpose of a Matrix; Symmetric Matrix

The matrix B is the **transpose** of the matrix A , written $B = A^T$, if each entry b_{ij} in B is the same as the entry a_{ji} in A , and conversely. If A is a matrix and if $A = A^T$, then the matrix A is **symmetric**.

EXAMPLE 9 Find A^T if

$$A = \begin{bmatrix} 1 & 4 & 5 \\ -3 & 2 & 7 \end{bmatrix}.$$

SOLUTION We have

$$A^T = \begin{bmatrix} 1 & -3 \\ 4 & 2 \\ 5 & 7 \end{bmatrix}.$$

Notice that the rows of A become the columns of A^T . $\blacksquare$

A symmetric matrix must be square. Symmetric matrices arise in some applications, as we shall see in Chapter 8.

EXAMPLE 10 Fill in the missing entries in the 4×4 matrix

$$\begin{bmatrix} 5 & -6 & & 8 \\ & 3 & & \\ -2 & 1 & 0 & 4 \\ & 11 & & -1 \end{bmatrix}$$

to make it symmetric.

SOLUTION Because rows must match corresponding columns, we obtain

$$\begin{bmatrix} 5 & -6 & -2 & 8 \\ -6 & 3 & 1 & 11 \\ -2 & 1 & 0 & 4 \\ 8 & 11 & 4 & -1 \end{bmatrix}.$$

In Example 10, note the symmetry in the main diagonal.

We have explained that we will often regard vectors in $\mathbb{R}^n$ as column vectors. If $\mathbf{a}$ and $\mathbf{b}$ are two column vectors in $\mathbb{R}^n$, the dot product $\mathbf{a} \cdot \mathbf{b}$ can be written in terms of the transpose operation and matrix multiplication—namely,

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^T \mathbf{b} = [a_1 \ a_2 \ \cdots \ a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}. \quad (7)$$

Strictly speaking, $\mathbf{a}^T \mathbf{b}$ is a 1×1 matrix, and its sole entry is $\mathbf{a} \cdot \mathbf{b}$. Identifying a 1×1 matrix with its sole entry should cause no difficulty. The use of Eq. (7) makes some formulas given later in the text much easier to handle.

Properties of Matrix Operations

For handy reference, we box the properties of matrix algebra and of the transpose operation. These properties are valid for all vectors, scalars, and matrices for which the indicated quantities are defined. The exercises ask for proofs of most of them. The proofs of the properties of matrix algebra not involving matrix multiplication are essentially the same as the proofs of the same properties presented for vector algebra in Section 1.1. We would expect this because those operations are performed just on corresponding entries, and every vector can be regarded as either a $1 \times n$ or an $n \times 1$ matrix.

Properties of Matrix Algebra

$A + B = B + A$	Commutative law of addition
$(A + B) + C = A + (B + C)$	Associative law of addition
$A + O = O + A = A$	Identity for addition
$r(A + B) = rA + rB$	A left distributive law
$(r + s)A = rA + sA$	A right distributive law
$(rs)A = r(sA)$	Associative law of scalar multiplication
$(rA)B = A(rB) = r(AB)$	Scalars pull through
$A(BC) = (AB)C$	Associative law of matrix multiplication
$IA = A$ and $BI = B$	Identity for matrix multiplication
$A(B + C) = AB + AC$	A left distributive law
$(A + B)C = AC + BC$	A right distributive law

Properties of the Transpose Operation

$(A^T)^T = A$	Transpose of the transpose
$(A + B)^T = A^T + B^T$	Transpose of a sum
$(AB)^T = B^T A^T$	Transpose of a product

EXAMPLE 11 Prove that $A(B + C) = AB + AC$ for any $m \times n$ matrix A and any $n \times s$ matrices B and C .

SOLUTION Let $A = [a_{ij}]$, $B = [b_{jk}]$ and $C = [c_{jk}]$. Note the use of j , which runs from 1 to n , as both the second index for entries in A and the first index for the entries in B and C . The entry in the i th row and k th column of $A(B + C)$ is

$$\sum_{j=1}^n a_{ij}(b_{jk} + c_{jk}).$$

By familiar properties of real numbers, this sum is also equal to

$$\sum_{j=1}^n (a_{ij}b_{jk} + a_{ij}c_{jk}) = \sum_{j=1}^n a_{ij}b_{jk} + \sum_{j=1}^n a_{ij}c_{jk}$$

which we recognize as the sum of the entries in the i th row and k th columns of the matrices AB and AC . This completes the proof. ■

SUMMARY

1. An $m \times n$ matrix is an ordered rectangular array of numbers containing m rows and n columns.
2. An $m \times 1$ matrix is a column vector with m components, and a $1 \times n$ matrix is a row vector with n components.
3. The product $A\mathbf{b}$ of an $m \times n$ matrix A and a column vector $\mathbf{b}$ with components $b_1, b_2, \dots, b_n$ is the column vector equal to the linear combination of the column vectors of A where the scalar coefficient of the j th column vector of A is b_j .
4. The product AB of an $m \times n$ matrix A and an $n \times s$ matrix B is the $m \times s$ matrix C whose j th column is A times the j th column of B . The entry c_{ij} in the i th row and j th column of C is the dot product of the i th row vector of A and the j th column vector of B . In general, $AB \neq BA$.
5. If $A = [a_{ij}]$ and $B = [b_{ij}]$ are matrices of the same size, then $A + B$ is the matrix of that size with entry $a_{ij} + b_{ij}$ in the i th row and j th column.
6. For any matrix A and scalar r , the matrix rA is found by multiplying each entry in A by r .
7. The transpose of an $m \times n$ matrix A is the $n \times m$ matrix A^T , which has as its k th row vector the k th column vector of A .
8. Properties of the matrix operations are given in boxed displays on page 45.

EXERCISES

In Exercises 1–16, let

$$A = \begin{bmatrix} -2 & 1 & 3 \\ 4 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & 1 & -2 \\ 5 & -1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & -1 \\ 0 & 6 \\ -3 & 2 \end{bmatrix}, \quad \text{and} \quad D = \begin{bmatrix} -4 & 2 \\ 3 & 5 \\ -1 & -3 \end{bmatrix}.$$

Compute the indicated quantity, if it is defined.

1. $3A$
2. $0B$
3. $A + B$
4. $B + C$
5. $C - D$
6. $4A - 2B$
7. AB
8. $(CD)^T$
9. $(2A)(5C)$
10. $(5D)(4B)$
11. A^2
12. $(AC)^2$
13. $(2A - B)D$
14. ADB
15. $(A^T)A$
16. BC and CB

17. Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- a. Find A^2 .
- b. Find A^T .

18. Let

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ 2 & 0 & 0 \end{bmatrix}.$$

- a. Find A^2 .
- b. Find A^T .

19. Consider the row and column vectors

$$\mathbf{x} = [-2, 3, -1] \text{ and } \mathbf{y} = \begin{bmatrix} 4 \\ -1 \\ 3 \end{bmatrix}.$$

Compute the matrix products $\mathbf{xy}$ and $\mathbf{yx}$.

20. Fill in the missing entries in the 4×4 matrix

$$\begin{bmatrix} 1 & -1 & & 5 \\ & 4 & & 8 \\ 2 & -7 & -1 & \\ & & 6 & 3 \end{bmatrix}$$

so that the matrix is symmetric.

21. Mark each of the following True or False. The statements involve matrices A , B , and C that are assumed to have appropriate size.
- ___ a. If $A = B$, then $AC = BC$.
 - ___ b. If $AC = BC$, then $A = B$.
 - ___ c. If $AB = O$, then $A = O$ or $B = O$.
 - ___ d. If $A + C = B + C$, then $A = B$.
 - ___ e. If $A^2 = I$, then $A = \pm I$.
 - ___ f. If $B = A^2$ and if A is $n \times n$ and symmetric, then $b_{ii} \geq 0$ for $i = 1, 2, \dots, n$.
 - ___ g. If $AB = C$ and if two of the matrices are square, then so is the third.
 - ___ h. If $AB = C$ and if C is a column vector, then so is B .
 - ___ i. If $A^2 = I$, then $A^n = I$ for all integers $n \geq 2$.
 - ___ j. If $A^2 = I$, then $A^n = I$ for all even integers $n \geq 2$.
22. a. Prove that, if A is a matrix and $\mathbf{x}$ is a row vector, then $\mathbf{x}A$ (if defined) is again a row vector.
- b. Prove that, if A is a matrix and $\mathbf{y}$ is a column vector, then $A\mathbf{y}$ (if defined) is again a column vector.
23. Let A be an $m \times n$ matrix and let $\mathbf{b}$ and $\mathbf{c}$ be column vectors with n components. Express the dot product $(A\mathbf{b}) \cdot (A\mathbf{c})$ as a product of matrices.
24. The product $A\mathbf{b}$ of a matrix and a column vector is equal to a linear combination of columns of A where the scalar coefficient of the j th column of A is b_j . In a similar fashion, describe the product $\mathbf{c}A$ of a row

vector $\mathbf{c}$ and a matrix A as a linear combination of vectors. [HINT: Consider $((\mathbf{c}A)^T)^T$.]

In Exercises 25–34, prove that the given relation holds for all vectors, matrices, and scalars for which the expressions are defined.

- 25. $A + B = B + A$
- 26. $(A + B) + C = A + (B + C)$
- 27. $(r + s)A = rA + sA$
- 28. $(rs)A = r(sA)$
- 29. $A(B + C) = AB + AC$
- 30. $(A^T)^T = A$
- 31. $(A + B)^T = A^T + B^T$
- 32. $(AB)^T = B^T A^T$
- 33. $(AB)C = A(BC)$
- 34. $(rA)B = A(rB) = r(AB)$
- 35. If B is an $m \times n$ matrix and if $B = A^T$, find the size of
 - a. A ,
 - b. AA^T ,
 - c. $A^T A$.
- 36. Let $\mathbf{v}$ and $\mathbf{w}$ be column vectors in $\mathbb{R}^n$. What is the size of $\mathbf{vw}^T$? What relationships hold between $\mathbf{vw}^T$ and $\mathbf{wv}^T$?
- 37. The Hilbert matrix H_n is the $n \times n$ matrix $[h_{ij}]$, where $h_{ij} = 1/(i + j - 1)$. Prove that the matrix H_n is symmetric.
- 38. Prove that, if A is a square matrix, then the matrix $A + A^T$ is symmetric.
- 39. Prove that, if A is a matrix, then the matrix AA^T is symmetric.
- 40. a. Prove that, if A is a square matrix, then $(A^2)^T = (A^T)^2$ and $(A^3)^T = (A^T)^3$. [HINT: Don't try to show that the matrices have equal entries; instead use Exercise 32.]
- b. State the generalization of part (a), and give a proof using mathematical induction (see Appendix A).
- 41. a. Let A be an $m \times n$ matrix, and let $\mathbf{e}_j$ be the $n \times 1$ column vector whose j th component is 1 and whose other components are 0. Show that $A\mathbf{e}_j$ is the j th column vector of A .

b. Let A and B be matrices of the same size.

i. Prove that, if $Ax = 0$ (the zero vector) for all x , then $A = O$, the zero matrix. [HINT: Use part (a).]

ii. Prove that, if $Ax = Bx$ for all x , then $A = B$. [HINT: Consider $A - B$.]

42. Let A and B be square matrices. Is

$$(A + B)^2 = A^2 + 2AB + B^2?$$

If so, prove it; if not, give a counterexample and state under what conditions the equation is true.

43. Let A and B be square matrices. Is

$$(A + B)(A - B) = A^2 - B^2?$$

If so, prove it; if not, give a counterexample and state under what conditions the equation is true.

44. An $n \times n$ matrix C is skew symmetric if $C^T = -C$. Prove that every square matrix A can be written uniquely as $A = B + C$ where B is symmetric and C is skew symmetric.

Matrix A commutes with matrix B if $AB = BA$.

45. Find all values of r for which

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & r \end{bmatrix} \text{ commutes with } \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

46. Find all values of r for which

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & r & 0 \\ 0 & 0 & 2 \end{bmatrix} \text{ commutes with } \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$



The software LINTEK includes a routine, MATCOMP, that performs the matrix operations described in this section. Let

$$A = \begin{bmatrix} 4 & 6 & 0 & 1 & -9 \\ 2 & 11 & 5 & 2 & -5 \\ -1 & 2 & -4 & 5 & 7 \\ 0 & 12 & -8 & 4 & 3 \\ 10 & 4 & 6 & 2 & -5 \end{bmatrix}$$

and

$$B = \begin{bmatrix} -8 & 15 & 4 & -11 \\ 3 & 5 & 6 & -2 \\ 0 & -1 & 12 & 5 \\ 1 & 13 & -15 & 7 \\ 6 & -8 & 0 & -5 \end{bmatrix}$$

Use MATCOMP in LINTEK to enter and store these matrices, and then compute the matrices in Exercises 47–54, if they are defined. Write down to hand in, if requested, the entry in the 3rd row, 4th column of the matrix.

47. $A^4 + A$ 50. BA^2 53. $(2A)^3 - A^5$
 48. A^2B 51. $B^T(2A)$ 54. $(A^T)^5$
 49. $A^3(A^T)^2$ 52. $AB(AB)^T$

MATLAB

To enter a matrix in MATLAB, start with a left bracket [and then type the entries across the rows, separating the entries by spaces and separating the rows by semicolons. Conclude with a right bracket]. To illustrate, we would enter the matrix

$$A = \begin{bmatrix} -1 & 5 \\ 13 & -4 \\ 7 & 0 \end{bmatrix} \text{ as } A = [-1 \ 5; 13 \ -4; 7 \ 0]$$

and MATLAB would then print it for us to proofread. Recall that to avoid having data printed again on the screen, we type a semicolon at the end of the data before pressing the Enter key. Thus if we enter

$$A = [-1 \ 5; 13 \ -4; 7 \ 0];$$

the matrix A will not be printed for proofreading. In MATLAB, we can enter

$A + B$ to find the sum

$A - B$ to find the difference

$A * B$ to find the product AB

$A ^ n$ to find the power A^n

$r*A$ to find the scalar multiple rA

A' to take the transpose of A

$\text{eye}(n)$ for the $n \times n$ identity matrix

$\text{zeros}(m,n)$ for an $m \times n$ matrix of 0's, or $\text{zeros}(n)$ if square

$\text{ones}(m,n)$ for an $m \times n$ matrix of 1's, or $\text{ones}(n)$ if square

$\text{rand}(m,n)$ for a matrix of random numbers from 0 to 1, or $\text{rand}(n)$ if square.

In MATLAB, $A(i, j)$ is the entry in the i th row and j th column of A , while $A(k)$ is the k th entry in A where entries are numbered consecutively starting at the upper left and proceeding down the first column, then down the second column, etc.

Access your MATLAB, and enter the matrices A and B given before Exercises 47–54. (We ask you to enter them manually this time to be sure you know how to enter matrices.) Proofread the data for A and B . If you find, for example, that you entered 6 rather than 5 for the entry in the 2nd row, 3rd column of A , you can correct your error by entering $A(2,3) = 5$.

- M1.** Exercises 47–54 are much easier to do with MATLAB than with LINTEK, because operations in LINTEK must be specified one at a time. Find the element in the 3rd row, 4th column of the given matrix.
- $B^T(2A)$
 - $AB(AB)^T$
 - $(2A)^3 - A^5$
- M2.** Enter $B(8)$. What answer did you get? Why did MATLAB give that answer?
- M3.** Enter `help :` to review the uses of the colon with matrices. Mastery of use of the colon is a real timesaver in MATLAB. Use the colon to set C equal to the 5×3 matrix consisting of the 3rd through the 5th columns of A . Then compute C^TC and write down your answer.
- M4.** Form a 5×9 matrix D whose first five columns are those of A and whose last four columns are those of B by
- entering $D = [A \ B]$, which works when A and B have the same number of rows,
 - first entering $D = A$ and then using the colon to specify that columns 6 through 9 of D are equal to B . Use the fact that $A(:, j)$ gives the j th column of A .
- Write down the entry in the 2nd row, 5th column of D^TD .
- M5.** Form a matrix E consisting of B with two rows of zeros put at the bottom and write down the entry in the 2nd row, 3rd column of E^TE by
- entering $Z = \text{zeros}(2,4)$ and then $E = [B; Z]$, which works when B and Z have the same number of columns,
 - first entering $E = B$ and then using the colon to specify that rows 6 through 7 of E are equal to Z . Use the fact that $A(i, :)$ gives the i th row of A .

- M6.** In mathematics, “mean” stands for “average,” so the mean of the numbers 2, 4, and 9 is their average $(2 + 4 + 9)/3 = 5$. In MATLAB, enter **help mean** to see what that function gives, and then enter **mean(A)**. Figure out a way to have MATLAB find the mean (average) of all 25 numbers in the matrix *A*. Find that mean and write down your answer.
- M7.** If *F* is a 10×10 matrix with random entries from 0 to 1, approximately what would you expect the mean value of those entries to be? Enter **help rand**, read what it says, and then generate such a matrix *F*. Using the idea in Exercise M6, compute the mean of the entries in *F*, and write down your answer. Repeat this several times.
- M8.** Enter **help ones** and read what it says. Write down a statement that you could enter in MATLAB to form from the matrix *F* of Exercise M7 a 10×10 matrix *G* that has random entries from -4 to 4 . Using the ideas in Exercise M6, find and write down the mean of the entries in the matrix *G*. Repeat this several times.
- M9.** In MATLAB, entering **mesh(X)** will draw a three-dimensional picture indicating the relative values of the entries in a matrix *X*, much as entering **plot(a)** draws a two-dimensional picture for the entries in a vector *a*. Enter **I = eye(16); mesh(I)** to see a graphic for the 16×16 identity matrix. Then enter **mesh(rot90(I))**. Enter **help rot90** and **help triu**. Enter **X = triu(ones(14)); mesh(X)**. Then enter **mesh(rot90(X))** and finally **mesh(rot90(X,-1))**.

MATLAB has the capability to draw surface graphs of a function $z = f(x, y)$ using the **mesh** function. This provides an excellent illustration of the use of a matrix to store data. As you experiment on your own, you may run out of computer memory, or try to form matrices larger than your MATLAB will accept. Entering **clear A** will erase a matrix *A* from memory to free up some space, and entering **clear** will erase all data previously entered. We suggest that you enter **clear** now before proceeding.

MATLAB can draw a surface of $z = f(x, y)$ over a rectangular region $a \leq x \leq b$, $c \leq y \leq d$ of the x, y -plane by computing values z of the function at points on a grid in the region. We can describe the region and the grid of points where we want values computed using the function **meshdom**. Review the use of the colon, using **help** : if you need to, and notice that entering **-3:1:3** will form a vector with first entry -3 and successive entries incremented by 1 until 3 is reached. If we enter

$$[X, Y] = \text{meshdom}(-3:1:3, -2.5:2.5)$$

then MATLAB will create two matrices *X* and *Y* containing, respectively, the x -coordinates and y -coordinates of a grid of points in the region $-3 \leq x \leq 3$ and $-2.5 \leq y \leq 2.5$. Because the x -increment is 1 and the y -increment is 0.5, we see that both *X* and *Y* will be 9×7 matrices.

Enter now

$$[X, Y] = \text{meshdom}(-3:1:3, -2.5:2.5); \quad (8)$$

and then enter **X** to see that matrix *X* and similarly view the matrix *Y*. We have specified where we want the function values computed.

Enter **help *** to recall that entering **A.*A** will produce the matrix whose entries are the squares of the entries in *A*. Thus entering **Z = X.*X + Y.*Y** in MATLAB will produce a matrix *Z* whose entry at a position corresponding to a grid point (x, y)

will be $x^2 + y^2$. Entering **mesh(Z)** will then create the mesh graph over the region $-3 \leq x \leq 3, -2 \leq y \leq 2$.

Enter now

$$Z = X .* X + Y .* Y; \quad (9)$$

$$\text{mesh}(Z) \quad (10)$$

to see this graph.

- M10. Using the up arrow, modify Eq. (8) to make both the x -increment and the y -increment 0.2. After pressing the Enter key, use the up arrow to get Eq. (9) and press the Enter key to form the larger matrix Z for these new grid points. Then create the mesh graph using Eq. (10).
- M11. Modify Eq. (9) and create the mesh graph for $z = x^2 - y^2$.
- M12. Change Eq. (8) so that the region will be $-3 \leq x \leq 3$ and $-3 \leq y \leq 3$ still with 0.2 for increments. Form the mesh graph for $z = 9 - x^2 - y^2$.
- M13. Mesh graphs for cylinders are especially nice. Draw the mesh graphs for
- $z = x^2$
 - $z = y^2$.
- M14. Change the mesh domain to $-4\pi \leq x \leq 4\pi, -3 \leq y \leq 3$ with x -increment 0.2 and y -increment 6. Recall that π can be entered as **pi**. Draw the mesh graphs for
- the cylinder $z = \sin(x)$,
 - the cylinder $z = x \sin(x)$, remembering to use **.***, and
 - the function $z = y \sin(x)$, which is not a cylinder but is pretty.

1.4

SOLVING SYSTEMS OF LINEAR EQUATIONS

As we have indicated, solving a system of linear equations is a fundamental problem of linear algebra. Many of the computational exercises in this text involve solving such linear systems. This section presents an algorithm for finding all solutions of any linear system.

The solution set of any system of equations is the intersection of the solution sets of the individual equations. That is, any solution of a system must be a solution of each equation in the system; and conversely, any solution of every equation in the system is considered to be a solution of the system. Bearing this in mind, we start with an intuitive discussion of the geometry of linear systems; a more detailed study of this geometry appears in Section 2.5.

The Geometry of Linear Systems

Frequently, students are under the impression that a linear system containing the same number of equations as unknowns always has a unique solution, whereas a system having more equations than unknowns never has a solution. The geometric interpretation of the problem shows that these statements are not true.

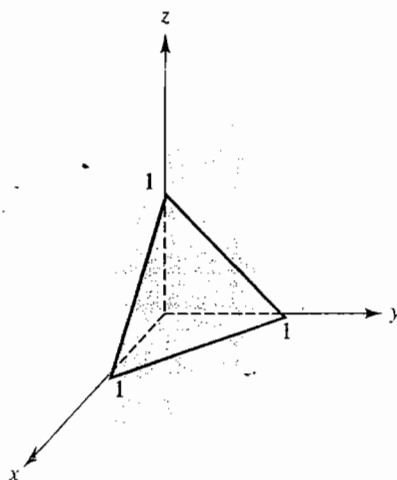


FIGURE 1.31
The plane $x + y + z = 1$.

We know that a single linear equation in two unknowns has a line in the plane as its solution set. Similarly, a single linear equation in three unknowns has a plane in space as its solution set. The solution set of $x + y + z = 1$ is the plane sketched in Figure 1.31. This geometric analysis can be extended to an equation that has more than three variables, but it is difficult for us to represent the solution set of such an equation graphically.

Two lines in the plane usually intersect at a single point; here the word *usually* means that, if the lines are selected in some random way, the chance

HISTORICAL NOTE SYSTEMS OF LINEAR EQUATIONS are found in ancient Babylonian and Chinese texts dating back well over 2000 years. The problems are generally stated in real-life terms, but it is clear that they are artificial and designed simply to train students in mathematical procedures. As an example of a Babylonian problem, consider the following, which has been slightly modified from the original found on a clay tablet from about 300 B.C.: There are two fields whose total area is 1800 square yards. One produces grain at a rate of $\frac{2}{3}$ bushel per square yard, the other at a rate of $\frac{1}{2}$ bushel per square yard. The total yield of the two fields is 1100 bushels. What is the size of each field? This problem leads to the system

$$x + y = 1800$$

$$\frac{2}{3}x + \frac{1}{2}y = 1100.$$

A typical Chinese problem, taken from the Han dynasty text *Nine Chapters of the Mathematical Art* (about 200 B.C.), reads as follows: There are three classes of corn, of which three bundles of the first class, two of the second, and one of the third make 39 measures. Two of the first, three of the second, and one of the third make 34 measures. And one of the first, two of the second, and three of the third make 26 measures. How many measures of grain are contained in one bundle of each class? The system of equations here is

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26.$$

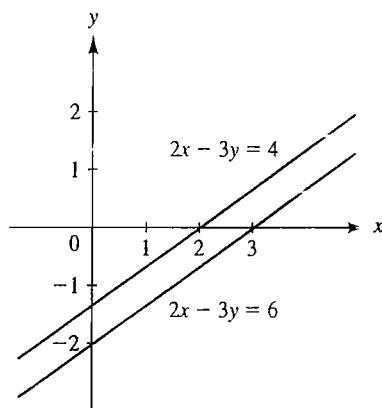


FIGURE 1.32
 $2x - 3y = 4$ is parallel to
 $2x - 3y = 6$.

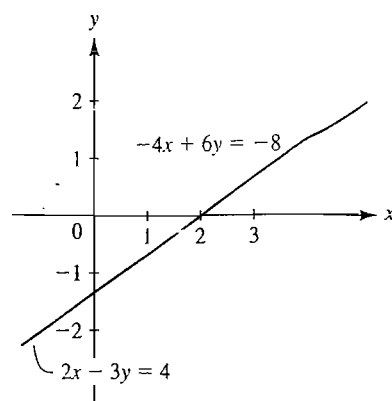


FIGURE 1.33
 $2x - 3y = 4$ and $-4x + 6y = -8$
 are the same line.

that they either are parallel (have empty intersection) or coincide (have an infinite number of points in their intersection) is very small. Thus we see that a system of two randomly selected equations in two unknowns can be expected to have a unique solution. However, it is *possible* for the system to have no solutions or an infinite number of solutions. For example, the equations

$$2x - 3y = 4$$

$$2x - 3y = 6$$

correspond to distinct parallel lines, as shown in Figure 1.32, and the system consisting of these equations has no solutions. Moreover, the equations

$$2x - 3y = 4$$

$$-4x + 6y = -8$$

correspond to the same line, as shown in Figure 1.33. All points on this line are solutions of this system of two equations. And because it is possible to have any number of lines in the plane—say, fifty lines—pass through a single point, it is possible for a system of fifty equations in only two unknowns to have a unique solution.

Similar illustrations can be made in space, where a linear equation has as its solution set a plane. Three randomly chosen planes can be expected to have a unique point in common. Two of them can be expected to intersect in a line (see Figure 1.34), which in turn can be expected to meet the third plane at a single point. However, it is possible for three planes to have no point in common, giving rise to a linear system with no solutions. It is also possible for all three planes to contain a common line, in which case the corresponding linear system will have an infinite number of solutions.

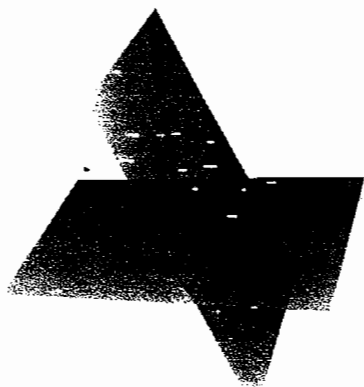


FIGURE 1.34
Two planes intersecting in a line.

Elementary Row Operations

We now describe operations that can be used to modify the equations of a linear system to obtain a system having the same solutions, but whose solutions are obvious. The most general type of linear system can have m equations in n unknowns. Such a system can be written as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned} \quad (1)$$

System (1) is completely determined by its $m \times n$ **coefficient matrix** $A = [a_{ij}]$ and by the column vector $\mathbf{b}$ with i th component b_i . The system can be written as the single matrix equation

$$A\mathbf{x} = \mathbf{b}, \quad (2)$$

where $\mathbf{x}$ is the column vector with i th component x_i . Any column vector $\mathbf{s}$ such that $A\mathbf{s} = \mathbf{b}$ is a **solution** of system (1).

The **augmented matrix** or **partitioned matrix**

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ & & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right] \quad (3)$$

is a shorthand summary of system (1). The coefficient matrix has been *augmented* by the column vector of constants. We denote matrix (3) by $[A | \mathbf{b}]$.

We shall see how to determine all solutions of system (1) by manipulating augmented matrix (3) using *elementary row operations*. The elementary row operations correspond to the following familiar operations with equations of system (1):

- R1 Interchange two equations in system (1).
- R2 Multiply an equation in system (1) by a nonzero constant.
- R3 Replace an equation in system (1) with the sum of itself and a multiple of a different equation of the system.

It is clear that operations R1 and R2 do not change the solution sets of the equations they affect. Therefore, they do not change the intersection of the solution sets of the equations; that is, the solution set of the system is unchanged. The fact that R2 does not change the solution set and the familiar algebraic principle, "Equals added to equals yield equals," show that any solution of both the i th and j th equations is also a solution of a new j th equation obtained by adding s times the i th equation to the j th equation. Thus operation R3 yields a system having all the solutions of the original one. Because the original system can be recovered from the new one by multiplying

HISTORICAL NOTE A MATRIX-REDUCTION METHOD of solving a system of linear equations occurs in the ancient Chinese work, *Nine Chapters of the Mathematical Art*. The author presents the following solution to the system

$$3x + 2y + z = 39$$

$$2x + 3y + z = 34$$

$$x + 2y + 3z = 26.$$

The diagram of the coefficients is to be set up on a "counting board":

$$\begin{array}{rrr} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 1 & 1 \\ 26 & 34 & 39 \end{array}$$

The author then instructs the reader to multiply the middle column by 3 and subsequently to subtract the right column "as many times as possible"; the same is to be done to the left column. The new diagrams are then

$$\begin{array}{rrr} 1 & 0 & 3 \\ 2 & 5 & 2 \\ 3 & 1 & 1 \\ 26 & 24 & 39 \end{array} \quad \text{and} \quad \begin{array}{rrr} 0 & 0 & 3 \\ 4 & 5 & 2 \\ 8 & 1 & 1 \\ 39 & 24 & 39 \end{array}$$

The next instruction is to multiply the left column by 5 and then to subtract the middle column as many times as possible. This gives

$$\begin{array}{rrr} 0 & 0 & 3 \\ 0 & 5 & 2 \\ 36 & 1 & 1 \\ 99 & 24 & 39 \end{array}$$

The system has thus been reduced to the system $3x + 2y + z = 39$, $5y + z = 24$, $36z = 99$, from which the complete solution is easily found.

the i th equation by $-s$ and adding it to the new j th equation (an R_3 operation), we see that the original system has all the solutions of the new one. Hence R_3 , too, does not alter the solution set of system (1).

These procedures applied to system (1) correspond to **elementary row operations** applied to augmented matrix (3). We list these in a box together with a suggestive notation for each.

Elementary Row Operations	Notations
(<i>Row interchange</i>) Interchange the i th and j th row vectors in a matrix.	$R_i \leftrightarrow R_j$
(<i>Row scaling</i>) Multiply the i th row vector in a matrix by a nonzero scalar s .	$R_i \rightarrow sR_i$
(<i>Row addition</i>) Add to the i th row vector of a matrix s times the j th row vector.	$R_i \rightarrow R_i + sR_j$

If a matrix B can be obtained from a matrix A by means of a sequence of elementary row operations, then A is **row equivalent** to B . Each elementary row operation can be undone by another of the same type. A row-addition operation $R_i \rightarrow R_i + sR_j$ can be undone by $R_i \rightarrow R_i - sR_j$. Row scaling, $R_i \rightarrow sR_i$ for $s \neq 0$, can be undone using $R_i \rightarrow (1/s)R_i$, while a row-interchange operation undoes itself. Thus, if B is row equivalent to A , then A is row equivalent to B ; we can simply speak of *row-equivalent matrices* A and B , which we denote by $A \sim B$. (See Exercise 55 in this regard.) We have just seen that the operations on a linear system $Ax = b$ corresponding to these elementary row operations on the augmented matrix $[A | b]$ do not change the solution set of the system. This gives us at once the following theorem, which is the foundation for the algorithm we will present for solving linear systems.

THEOREM 1.6 Invariance of Solution Sets Under Row Equivalence

If $[A | b]$ and $[H | c]$ are row-equivalent augmented matrices, then the linear systems $Ax = b$ and $Hx = c$ have the same solution sets.

Row-Echelon Form

We will solve a linear system $Ax = b$ by row-reducing the augmented matrix $[A | b]$ to an augmented matrix $[H | c]$, where H is a matrix in *row-echelon form* (which we now define).

DEFINITION 1.12 Row-Echelon Form, Pivot

A matrix is in **row-echelon form** if it satisfies two conditions:

1. All rows containing only zeros appear below rows with nonzero entries.
2. The first nonzero entry in any row appears in a column to the right of the first nonzero entry in any preceding row.

For such a matrix, the first nonzero entry in a row is the **pivot** for that row.

EXAMPLE 1 Determine which of the matrices

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 4 & 0 \\ 1 & 3 & 2 \\ 0 & 0 & 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & -1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & 3 & 2 & 5 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

are in row-echelon form.

SOLUTION Matrix A is not in row-echelon form, because the second row (consisting of all zero entries) is not below the third row (which has a nonzero entry).

Matrix B is not in row-echelon form, because the first nonzero entry in the second row does not appear in a column to the right of the first nonzero entry in the first row.

Matrix C is in row-echelon form, because both conditions of Definition 1.12 are satisfied. The pivots are -1 and 3 .

Matrix D satisfies both conditions as well, and is in row-echelon form. The pivots are the entries 1 . ■

Solutions of $Hx = c$

We illustrate by examples that, if a linear system $Hx = c$ has coefficient matrix H in row-echelon form, it is easy to determine all solutions of the system. We color the pivots in H in these examples.

EXAMPLE 2 Find all solutions of $Hx = c$, where

$$[H | c] = \left[\begin{array}{ccc|c} -5 & -1 & 3 & 3 \\ 0 & 3 & 5 & 8 \\ 0 & 0 & 2 & -4 \end{array} \right].$$

SOLUTION The equations corresponding to this augmented matrix are

$$\begin{aligned} -5x_1 - x_2 + 3x_3 &= 3 \\ 3x_2 + 5x_3 &= 8 \\ 2x_3 &= -4. \end{aligned}$$

From the last equation, we obtain $x_3 = -2$. Substituting into the second equation, we have

$$3x_2 + 5(-2) = 8, \quad 3x_2 = 18, \quad x_2 = 6.$$

Finally, we substitute these values for x_2 and x_3 into the top equation, obtaining

$$-5x_1 - 6 + 3(-2) = 3, \quad -5x_1 = 15, \quad x_1 = -3.$$

Thus the only solution is

$$\mathbf{x} = \begin{bmatrix} -3 \\ 6 \\ -2 \end{bmatrix},$$

or equivalently, $x_1 = -3$, $x_2 = 6$, $x_3 = -2$. ■

The procedure for finding the solution of $H\mathbf{x} = \mathbf{c}$ illustrated in Example 2 is called **back substitution**, because the values of the variables are found in backward order, starting with the variable with the largest subscript.

By multiplying each nonzero row in $[H | \mathbf{b}]$ by the reciprocal of its pivot, we can assume that each pivot in H is 1. We assume that this is the case in the next two examples.

EXAMPLE 3 Use back substitution to find all solutions of $H\mathbf{x} = \mathbf{c}$, where

$$[H | \mathbf{c}] = \left[\begin{array}{ccc|c} 1 & -3 & 5 & 3 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & -1 \end{array} \right].$$

SOLUTION The equation corresponding to the last row of this augmented matrix is

$$0x_1 + 0x_2 + 0x_3 = -1.$$

This equation has no solutions, because the left side is 0 for any values of the variables and the right side is -1 . ■

DEFINITION 1.13 Consistent Linear System

A linear system having no solutions is **inconsistent**. If it has one or more solutions, the linear system is said to be **consistent**.

Now we illustrate a many-solutions case.

EXAMPLE 4 Use back substitution to find all solutions of $H\mathbf{x} = \mathbf{c}$, where

$$[H | \mathbf{c}] = \left[\begin{array}{ccccc|c} 1 & -3 & 0 & 5 & 0 & 4 \\ 0 & 0 & 1 & 2 & 0 & -7 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right].$$

SOLUTION The linear system corresponding to this augmented matrix is

$$\begin{aligned} x_1 - 3x_2 + 5x_4 &= 4 \\ x_3 + 2x_4 &= -7 \\ x_5 &= 1. \end{aligned}$$

We solve each equation for the variable corresponding to the colored pivot in the matrix. Thus we obtain

$$\begin{aligned} x_1 &= 3x_2 - 5x_4 + 4 \\ x_3 &= -2x_4 - 7 \\ x_5 &= 1. \end{aligned} \tag{4}$$

Notice that x_2 and x_4 correspond to columns of H containing no pivot. We can assign any value r we please to x_2 and any value s to x_4 , and we can then use system (4) to determine corresponding values for x_1 , x_3 , and x_5 . Thus the system has an infinite number of solutions. We describe all solutions by the vector equation

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 3r - 5s + 4 \\ r \\ -2s - 7 \\ s \\ 1 \end{bmatrix} \quad \text{for any scalars } r \text{ and } s. \tag{5}$$

We call x_2 and x_4 **free variables**, and we refer to Eq. (5) as the **general solution** of the system. We obtain **particular solutions** by setting r and s equal to specific values. For example, we obtain

$$\begin{bmatrix} 2 \\ 1 \\ -9 \\ 1 \\ 1 \end{bmatrix} \text{ for } r = s = 1 \quad \text{and} \quad \begin{bmatrix} 25 \\ 2 \\ -1 \\ -3 \\ 1 \end{bmatrix} \text{ for } r = 2, s = -3. \quad \blacksquare$$

Gauss Reduction of $A\mathbf{x} = \mathbf{b}$ to $H\mathbf{x} = \mathbf{c}$

We now show how to reduce an augmented matrix $[A | \mathbf{b}]$ to $[H | \mathbf{c}]$, where H is in row-echelon form, using a sequence of elementary row operations. Examples 2 through 4 illustrated how to use back substitution afterward to find solutions of the system $H\mathbf{x} = \mathbf{c}$, which are the same as the solutions of $A\mathbf{x} = \mathbf{b}$, by Theorem 1.6. This procedure for solving $A\mathbf{x} = \mathbf{b}$ is known as **Gauss**

reduction with back substitution. In the box below, we give an outline for reducing a matrix A to row-echelon form.

Reducing a Matrix A to Row-Echelon Form H

1. If the first column of A contains only zero entries, cross it off mentally. Continue in this fashion until the left column of the remaining matrix has a nonzero entry or until the columns are exhausted.
2. Use row interchange, if necessary, to obtain a nonzero entry (pivot) p in the top row of the first column of the remaining matrix. For each row below that has a nonzero entry r in the first column, add $-r/p$ times the top row to that row to create a zero in the first column. In this fashion, create zeros below p in the entire first column of the remaining matrix.
3. Mentally cross off this first column and the first row of the matrix, to obtain a smaller matrix. (See the shaded portion of the third matrix in the solution of Example 5.) Go back to step 1, and repeat the process with this smaller matrix until either no rows or no columns remain.

EXAMPLE 5 Reduce the matrix

$$\begin{bmatrix} 2 & -4 & 2 & -2 \\ 2 & -4 & 3 & -4 \\ 4 & -8 & 3 & -2 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

to row-echelon form, making all pivots 1.

SOLUTION We follow the boxed outline and color the pivots of 1. Remember that the symbol $\sim$ denotes row-equivalent matrices.

$$\begin{bmatrix} 2 & -4 & 2 & -2 \\ 2 & -4 & 3 & -4 \\ 4 & -8 & 3 & -2 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

Multiply the first row by $\frac{1}{2}$, to produce a pivot of 1 in the next matrix.

$$R_1 \rightarrow \frac{1}{2}R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & -1 \\ 2 & -4 & 3 & -4 \\ 4 & -8 & 3 & -2 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

Add -2 times row 1 to row 2, and then add -4 times row 1 to row 3, to obtain the next matrix.

$$R_2 \rightarrow R_2 - 2R_1; \quad R_3 \rightarrow R_3 - 4R_1$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

Cross off the first shaded column of zeros (mentally), to obtain the next shaded matrix.

$$\sim \begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & -1 & 2 \\ 0 & 0 & -1 & 2 \end{bmatrix} \quad \begin{array}{l} \text{Add row 2 to rows 3 and 4, to obtain the final} \\ \text{matrix.} \end{array}$$

$$R_3 \rightarrow R_3 + 1R_2; \quad R_4 \rightarrow R_4 + 1R_2$$

$$\sim \begin{bmatrix} 1 & -2 & 1 & -1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

This last matrix is row-echelon form, with both pivots equal to 1. ■

To solve a linear system $Ax = b$, we form the augmented matrix $[A | b]$ and row-reduce it to $[H | c]$, where H is in row-echelon form. We can follow the steps outlined in the box preceding Example 5 for row-reducing A to H . Of course, we always perform the elementary row operations on the full augmented matrix, including the entries in the column to the right of the partition.

EXAMPLE 6 Solve the linear system

$$\begin{aligned} x_2 - 3x_3 &= -5 \\ 2x_1 + 3x_2 - x_3 &= 7 \\ 4x_1 + 5x_2 - 2x_3 &= 10. \end{aligned}$$

SOLUTION We reduce the corresponding augmented matrix, using elementary row operations. Pivots are colored.

$$\begin{aligned} \left[\begin{array}{ccc|c} 0 & 1 & -3 & -5 \\ 2 & 3 & -1 & 7 \\ 4 & 5 & -2 & 10 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 4 & 5 & -2 & 10 \end{array} \right] & R_1 \leftrightarrow R_2 \\ \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 4 & 5 & -2 & 10 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 0 & -1 & 0 & -4 \end{array} \right] & R_3 \rightarrow R_3 - 2R_1 \\ \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 0 & -1 & 0 & -4 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & -3 & -9 \end{array} \right] & R_3 \rightarrow R_3 + 1R_2 \end{aligned}$$

HISTORICAL NOTE THE GAUSS SOLUTION METHOD is so named because Gauss described it in a paper detailing the computations he made to determine the orbit of the asteroid Pallas. The parameters of the orbit had to be determined by observations of the asteroid over a 6-year period from 1803 to 1809. These led to six linear equations in six unknowns with quite complicated coefficients. Gauss showed how to solve these equations by systematically replacing them with a new system in which only the first equation had all six unknowns, the second equation included five unknowns, the third equation only four, and so on, until the sixth equation had but one. This last equation could, of course, be easily solved; the remaining unknowns were then found by back substitution.

From the last augmented matrix, we could proceed to write the corresponding equations (as in Example 2) and to solve in succession for x_3 , x_2 , and x_1 by back substitution. However, it makes sense to keep using our shorthand, without writing out variables, and to do our back substitution in terms of augmented matrices. Starting with the final augmented matrix in the preceding set, we obtain

$$\begin{aligned} \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & -3 & -9 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & 1 & 3 \end{array} \right] & \begin{array}{l} R_3 \rightarrow -\frac{1}{3}R_3 \\ \text{(This shows that } x_3 = 3.) \end{array} \\ \\ \left[\begin{array}{ccc|c} 2 & 3 & -1 & 7 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & 1 & 3 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 2 & 3 & 0 & 10 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right] & \begin{array}{l} R_1 \rightarrow R_1 + 1R_3; \quad R_2 \rightarrow R_2 + 3R_3 \\ \text{(This shows that } x_2 = 4.) \end{array} \\ \\ \left[\begin{array}{ccc|c} 2 & 3 & 0 & 10 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right] &\sim \left[\begin{array}{ccc|c} 2 & 0 & 0 & -2 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right] & \begin{array}{l} R_1 \rightarrow R_1 - 3R_2 \\ \text{(This shows that } x_1 = -1.) \end{array} \end{aligned}$$

We have found the solution: $\mathbf{x} = [x_1, x_2, x_3] = [-1, 4, 3]$. ■

In Example 6, we had to write down as many matrices to execute the back substitution as we wrote to reduce the original augmented matrix to row-echelon form. We can avoid this by creating zeros *above* as well as below each pivot as we reduce the matrix to row-echelon form. This is known as the **Gauss–Jordan method**. We show in Chapter 10 that, for a large system, it takes about 50% more time for a computer to use the Gauss–Jordan method than to use the Gauss method with back substitution illustrated in Example 6; there are actually about 50% more arithmetic operations involved. Creating the zeros above the pivots requires less computation if we do it *after* the matrix is reduced to row-echelon form than if we do it as we go along. However, when one is working with pencil and paper, fixing up a whole column in a single step avoids writing so many matrices. Our next example illustrates the Gauss–Jordan procedure.

EXAMPLE 7 Determine whether the vector $\mathbf{b} = [1, -7, -4]$ is in the span of the vectors $\mathbf{v} = [2, 1, 1]$ and $\mathbf{w} = [1, 3, 2]$.

SOLUTION We know that $\mathbf{b}$ is in $\text{sp}(\mathbf{v}, \mathbf{w})$ if and only if $\mathbf{b} = x_1\mathbf{v} + x_2\mathbf{w}$ for some scalars x_1 and x_2 . This vector equation is equivalent to the linear system

$$\begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -7 \\ -4 \end{bmatrix}.$$

Reducing the appropriate augmented matrix, we obtain

$$\left[\begin{array}{cc|c} 2 & 1 & 1 \\ 1 & 3 & -7 \\ 1 & 2 & -4 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 3 & -7 \\ 2 & 1 & 1 \\ 1 & 2 & -4 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 3 & -7 \\ 0 & -5 & 15 \\ 0 & -1 & 3 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array} \right].$$

$R_1 \leftrightarrow R_2$ $R_2 \rightarrow R_2 - 2R_1$ $R_2 \rightarrow \frac{1}{5}R_2$
 (to avoid $R_3 \rightarrow R_3 - 1R_1$ $R_1 \rightarrow R_1 - 3R_2$
 fractions) $R_3 \rightarrow R_3 + 1R_2$

The left side of the final augmented matrix is in *reduced row-echelon form*. From the solution $x_1 = 2$ and $x_2 = -3$, we see that $\mathbf{b} = 2\mathbf{v} - 3\mathbf{w}$, which is indeed in $\text{sp}(\mathbf{v}, \mathbf{w})$. ■

The linear system $A\mathbf{x} = \mathbf{b}$ displayed in Eq. (1) can be written in the form

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

This equation expresses a typical column vector $\mathbf{b}$ in $\mathbb{R}^m$ as a linear combination of the column vectors of the matrix A if and only if scalars $x_1, x_2, \dots, x_n$ can be found to satisfy that equation. Example 7 illustrates this. We phrase this result as follows:

Let A be an $m \times n$ matrix. The linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if the vector $\mathbf{b}$ in $\mathbb{R}^m$ is in the span of the column vectors of A .

A matrix in row-echelon form with all pivots equal to 1 and with zeros above as well as below each pivot is said to be in **reduced row-echelon form**. Thus the Gauss–Jordan method consists of using elementary row operations on an augmented matrix $[A \mid \mathbf{b}]$ to bring the coefficient matrix A into reduced

HISTORICAL NOTE THE JORDAN HALF OF THE GAUSS–JORDAN METHOD is essentially a systematic technique of back substitution. In this form, it was first described by Wilhelm Jordan (1842–1899), a German professor of geodesy, in the third (1888) edition of his *Handbook of Geodesy*. Although Jordan’s arrangement of his calculations is different from the one presented here, partly because he was always applying the method to the symmetric system of equations arising out of a least-squares application in geodesy (see Section 6.5), Jordan’s method uses the same arithmetic and arrives at the same answers for the unknowns.

Wilhelm Jordan was prominent in his field in the late nineteenth century, being involved in several geodetic surveys in Germany and in the first major survey of the Libyan desert. He was the founding editor of the German geodesy journal and was widely praised as a teacher of his subject. His interest in finding a systematic method of solving large systems of linear equations stems from their frequent appearance in problems of triangulation.

row-echelon form. It can be shown that the reduced row-echelon form of a matrix A is unique. (See Section 2.3, Exercise 33.)

The examples we have given illustrate the three possibilities for solutions of a linear system—namely, no solutions (inconsistent system), a unique solution, or an infinite number of solutions. We state this formally in a theorem and prove it.

THEOREM 1.7 Solutions of $Ax = b$

Let $Ax = b$ be a linear system, and let $[A | b] \sim [H | c]$, where H is in row-echelon form.

1. The system $Ax = b$ is inconsistent if and only if the augmented matrix $[H | c]$ has a row with all entries 0 to the left of the partition and a nonzero entry to the right of the partition.
2. If $Ax = b$ is consistent and every column of H contains a pivot, the system has a unique solution.
3. If $Ax = b$ is consistent and some column of H has no pivot, the system has infinitely many solutions, with as many free variables as there are pivot-free columns in H .

PROOF If $[H | c]$ has an i th row with all entries 0 to the left of the partition and a nonzero entry c_i to the right of the partition, the corresponding i th equation in the system $Hx = c$ is $0x_1 + 0x_2 + \cdots + 0x_n = c_i$, which has no solutions; therefore, the system $Ax = b$ has no solutions, by Theorem 1.6. The next paragraph shows that, if H contains no such row, we can find a solution to the system. Thus the system is inconsistent if and only if H contains such a row.

Assume now that $[H | c]$ has no row with all entries 0 to the left of the partition and a nonzero entry to the right. If the i th row of $[H | c]$ is a zero row vector, the corresponding equation $0x_1 + 0x_2 + \cdots + 0x_n = 0$ is satisfied for all values of the variables x_j , and thus it can be deleted from the system $Hx = c$. Assume that this has been done wherever possible, so that $[H | c]$ has no zero row vectors. For each j such that the j th column has no pivot, we can set x_j equal to any value we please (as in Example 4) and then, starting from the last remaining equation of the system and working back to the first, solve in succession for the variables corresponding to the columns containing the pivots. If some column j has no pivot, there are an infinite number of solutions, because x_j can be set equal to any value. On the other hand, if every column has a pivot (as in Examples 2, 6, and 7), the value of each x_j is uniquely determined. $\blacktriangle$

With reference to item (3) of Theorem 1.7, the number of free variables in the solution set of a system $Ax = b$ depends only on the system, and not on the

way in which the matrix A is reduced to row-echelon form. This follows from the uniqueness of the reduced row-echelon form. (See Exercise 33 in Section 2.3.)

Elementary Matrices

The elementary row operations we have performed can actually be carried out by means of matrix multiplication. Although it is not efficient to row-reduce a matrix by multiplying it by other matrices, representing row reduction as a product of matrices is a useful theoretical tool. For example, we use elementary matrices in Section 1.5 to show that, for square matrices A and C , if $AC = I$, then $CA = I$. We use them again in Section 4.2 to demonstrate the multiplicative property of determinants, and again in Section 10.2 to exhibit a factorization of some square matrices A into a product LU of a lower-triangular matrix L and an upper-triangular matrix U .

If we interchange its second and third rows, the 3×3 identity matrix

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ becomes } E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

If $A = [a_{ij}]$ is a 3×3 matrix, we can compute EA , and we find that

$$EA = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{31} & a_{32} & a_{33} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}.$$

We have interchanged the second and third rows of A by multiplying A on the left by E .

DEFINITION 1.14 Elementary Matrix

Any matrix that can be obtained from an identity matrix by means of one elementary row operation is an **elementary matrix**.

We leave the proof of the following theorem as Exercises 52 through 54.

THEOREM 1.8 Use of Elementary Matrices

Let A be an $m \times n$ matrix, and let E be an $m \times m$ elementary matrix. Multiplication of A on the left by E effects the same elementary row operation on A that was performed on the identity matrix to obtain E .

Thus row reduction of a matrix to row-echelon form can be accomplished by successive multiplication on the left by elementary matrices. In other words, if A can be reduced to H through elementary row operations, there exist elementary matrices $E_1, E_2, \dots, E_t$ such that

$$H = (E_t \cdots E_2 E_1)A.$$

Again, this is by no means an efficient way to execute row reduction, but such an algebraic representation of H in terms of A is sometimes handy in proving theorems.

EXAMPLE 8 Let

$$A = \begin{bmatrix} 0 & 1 & -3 \\ 2 & 3 & -1 \\ 4 & 5 & -2 \end{bmatrix}.$$

Find a matrix C such that CA is a matrix in row-echelon form that is row equivalent to A .

SOLUTION We row reduce A to row-echelon form H and write down, for each row operation, the elementary matrix obtained by performing the same operation on the 3×3 identity matrix.

Reduction of A	Row Operation	Elementary Matrix
$A = \begin{bmatrix} 0 & 1 & -3 \\ 2 & 3 & -1 \\ 4 & 5 & -2 \end{bmatrix}$	$R_1 \leftrightarrow R_2$	$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
$\sim \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & -3 \\ 4 & 5 & -2 \end{bmatrix}$	$R_3 \rightarrow R_3 - 2R_1$	$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$
$\sim \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & -3 \\ 0 & -1 & 0 \end{bmatrix}$	$R_3 \rightarrow R_3 + 1R_2$	$E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$
$\sim \begin{bmatrix} 2 & 3 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & -3 \end{bmatrix} = H.$		

Thus, we must have $E_3(E_2(E_1A)) = H$; so the desired matrix C is

$$C = E_3 E_2 E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix}.$$

To compute C , we do not actually have to multiply out $E_3 E_2 E_1$. We know that multiplication of E_1 on the left by E_2 simply adds -2 times row 1 of E_1 to its

row 3, and subsequent multiplication on the left by E_3 adds row 2 to row 3 of the matrix E_2E_1 . Thus we can find C by executing the same row-reduction steps on I that we executed to change A to H —namely,

$$\begin{array}{ccccccc} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \sim & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \sim & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} & \sim & \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & -2 & 1 \end{bmatrix} \\ I & & E_1 & & E_2E_1 & & C = E_3E_2E_1 \end{array}$$

We can execute analogous *elementary column* operations on a matrix by multiplying the matrix on the *right* by an elementary matrix. Column reduction of a matrix A is not important for us in this chapter, because it does not preserve the solution set of $Ax = b$ when applied to the augmented matrix $[A | b]$. However, we will have occasion to refer to column reduction when computing determinants. The effect of multiplication of a matrix A on the right by elementary matrices is explored in Exercises 36–38 in Section 1.5.

SUMMARY

1. A linear system has an associated augmented (or partitioned) matrix, having the coefficient matrix of the system on the left of the partition and the column vector of constants on the right of the partition.
2. The elementary row operations on a matrix are as follows:
 - (Row interchange) Interchange of two rows,
 - (Row scaling) Multiplication of a row by a nonzero scalar;
 - (Row addition) Addition of a multiple of a row to a different row.
3. Matrices A and B are row equivalent (written $A \sim B$) if A can be transformed into B by a sequence of elementary row operations.
4. If $Ax = b$ and $Hx = c$ are systems such that the augmented matrices $[A | b]$ and $[H | c]$ are row equivalent, the systems $Ax = b$ and $Hx = c$ have the same solution set.
5. A matrix is in row-echelon form if:
 - a. All rows containing only zero entries are grouped together at the bottom of the matrix.
 - b. The first nonzero element (the pivot) in any row appears in a column to the right of the first nonzero element in any preceding row.
6. A matrix is in reduced row-echelon form if it is in row-echelon form and, in addition, each pivot is 1 and is the only nonzero element in its column. Every matrix is row equivalent to a unique matrix in reduced row-echelon form.
7. In the Gauss method with back substitution, we solve a linear system by reducing the augmented matrix so that the portion to the left of the

partition is in row-echelon form. The solution is then found by back substitution.

8. The Gauss-Jordan method is similar to the Gauss method, except that pivots are adjusted to be 1 and zeros are created above as well as below the pivots.
9. A linear system $Ax = b$ has no solutions if and only if, after $[A | b]$ is row-reduced so that A is transformed into row-echelon form, there exists a row with only zero entries to the left of the partition but with a nonzero entry to the right of the partition. The linear system is then *inconsistent*.
10. If $Ax = b$ is a consistent linear system and if a row-echelon form H of A has at least one column containing no (nonzero) pivot, the system has an infinite number of solutions. The free variables corresponding to the columns containing no pivots can be assigned any values, and the reduced linear system can then be solved for the remaining variables.
11. An elementary matrix E is one obtained by applying a single elementary row operation to an identity matrix I . Multiplication of a matrix A on the left by E effects the same elementary row operation on A .

EXERCISES

In Exercises 1–6, reduce the matrix to (a) row-echelon form, and (b) reduced row-echelon form. Answers to (a) are not unique, so your answer may differ from the one at the back of the text.

1.
$$\begin{bmatrix} 2 & 1 & 4 \\ 1 & 3 & 2 \\ 3 & -1 & 6 \end{bmatrix}$$

2.
$$\begin{bmatrix} 2 & 4 & -2 \\ 4 & 8 & 3 \\ -1 & -3 & 0 \end{bmatrix}$$

3.
$$\begin{bmatrix} 0 & 2 & -1 & 3 \\ -1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 3 \\ 1 & 5 & 5 & 9 \end{bmatrix}$$

4.
$$\begin{bmatrix} 0 & 0 & 3 & -2 \\ 0 & 0 & 1 & 2 \\ 1 & 3 & 2 & -4 \end{bmatrix}$$

5.
$$\begin{bmatrix} -1 & 3 & 0 & 1 & 4 \\ 1 & -3 & 0 & 0 & -1 \\ 2 & -6 & 2 & 4 & 0 \\ 0 & 0 & 1 & 3 & -4 \end{bmatrix}$$

6.
$$\begin{bmatrix} 0 & 0 & 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 & -1 & 3 \\ 2 & 4 & -1 & 3 & 2 & -1 \end{bmatrix}$$

In Exercises 7–12, describe all solutions of a linear system whose corresponding augmented matrix can be row-reduced to the given matrix. If requested, also give the indicated particular solution, if it exists.

7.
$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 1 & 4 & 2 \end{array} \right], \text{ solution with } x_3 = 2$$

8.
$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 2 & 4 \end{array} \right]$$

9.
$$\left[\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right],$$

solution with $x_3 = 3, x_4 = -2$

10.
$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 3 & 0 & -4 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right],$$

solution with $x_2 = 2, x_3 = 1$

11.
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 1 & 0 & -5 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$$